금융투자론

파생금융상품

3. 스왑(swaps)

Nature of Swaps

A swap is **an agreement to exchange cash flows** at specified future times according to certain specified rules.

An Example of a "Plain Vanilla" Interest Rate Swap

- An agreement by Microsoft to receive 6-month LIBOR & pay a fixed rate of 5% per annum every 6 months for 3 years on a notional principal of $100 million
- Next slide illustrates cash flows

Cash Flows to Microsoft

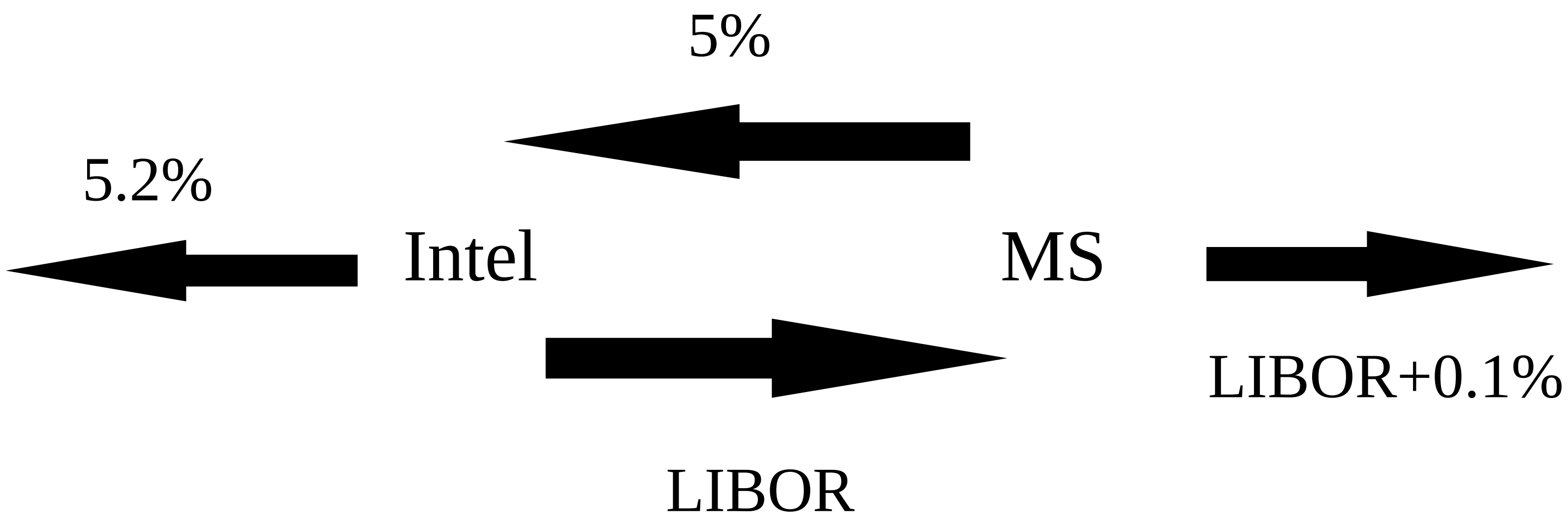
		---------Millions of Dollars---------		
	LIBOR	*FLOATING*	*FIXED*	Net
Date	Rate	Cash Flow	Cash Flow	Cash Flow
Mar.5, 2007	4.2%			
Sept. 5, 2007	4.8%	+2.10	–2.50	–0.40
Mar.5, 2008	5.3%	+2.40	–2.50	–0.10
Sept. 5, 2008	5.5%	+2.65	–2.50	+0.15
Mar.5, 2009	5.6%	+2.75	–2.50	+0.25
Sept. 5, 2009	5.9%	+2.80	–2.50	+0.30
Mar.5, 2010	6.4%	+2.95	–2.50	+0.45

Typical Uses of an Interest Rate Swap

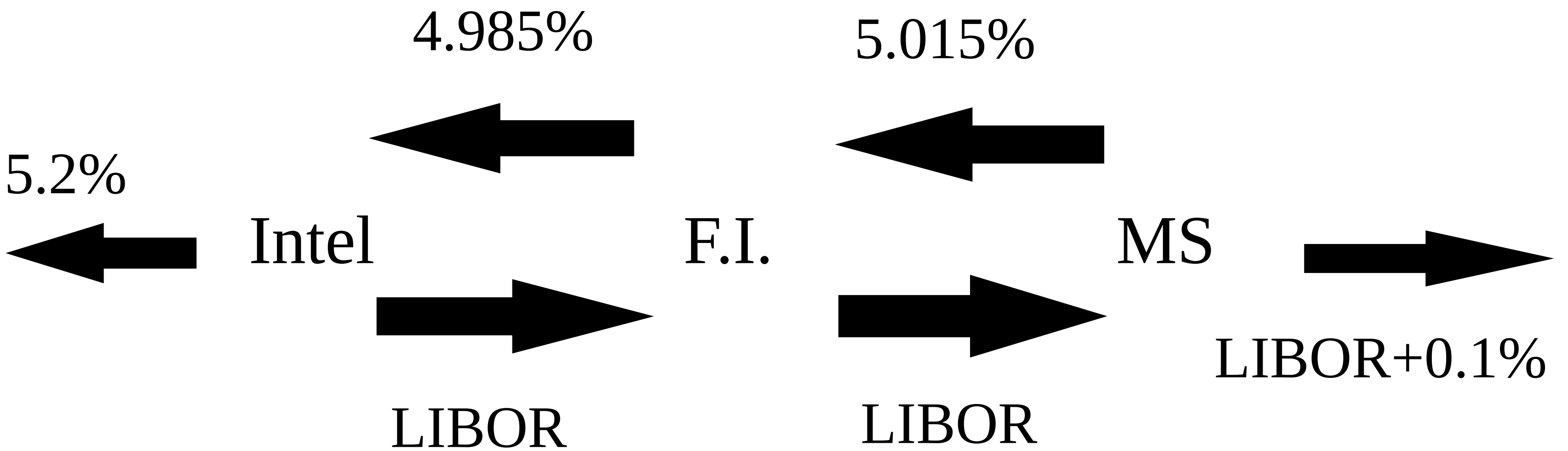
- **Converting a liability** from
  - fixed rate to floating rate
  - floating rate to fixed rate

- **Converting an investment** from
  - fixed rate to floating rate
  - floating rate to fixed rate

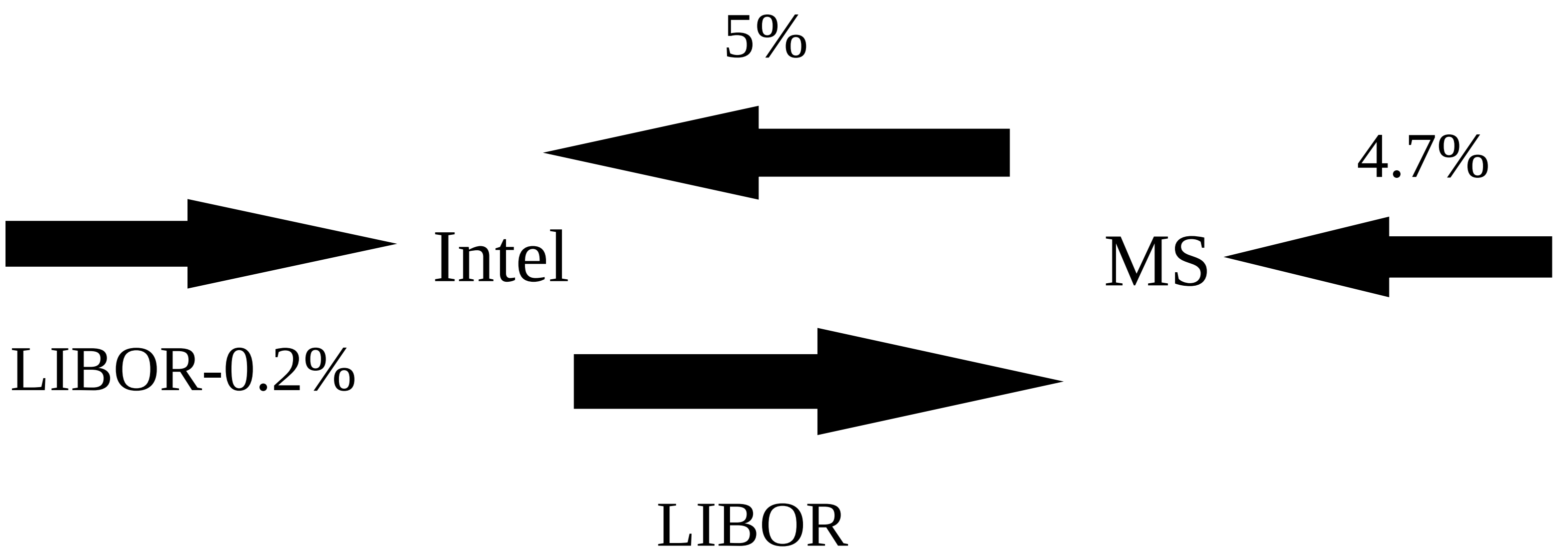
Intel and Microsoft (MS) Transform a Liability



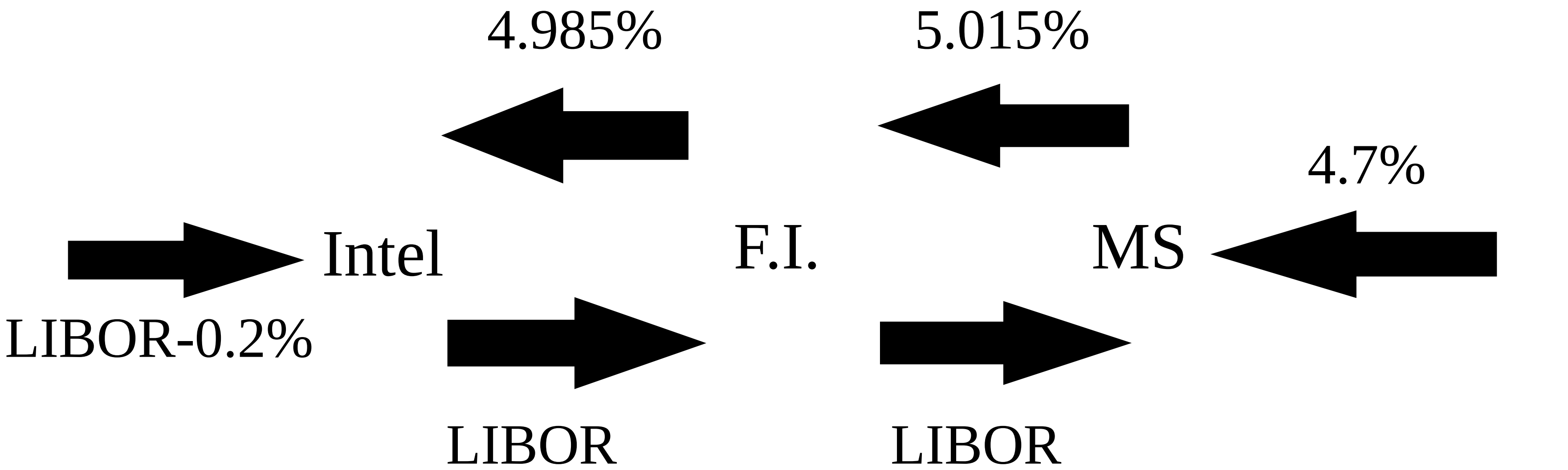
Financial Institution is Involved



Intel and Microsoft (MS) Transform an Asset



Financial Institution is Involved

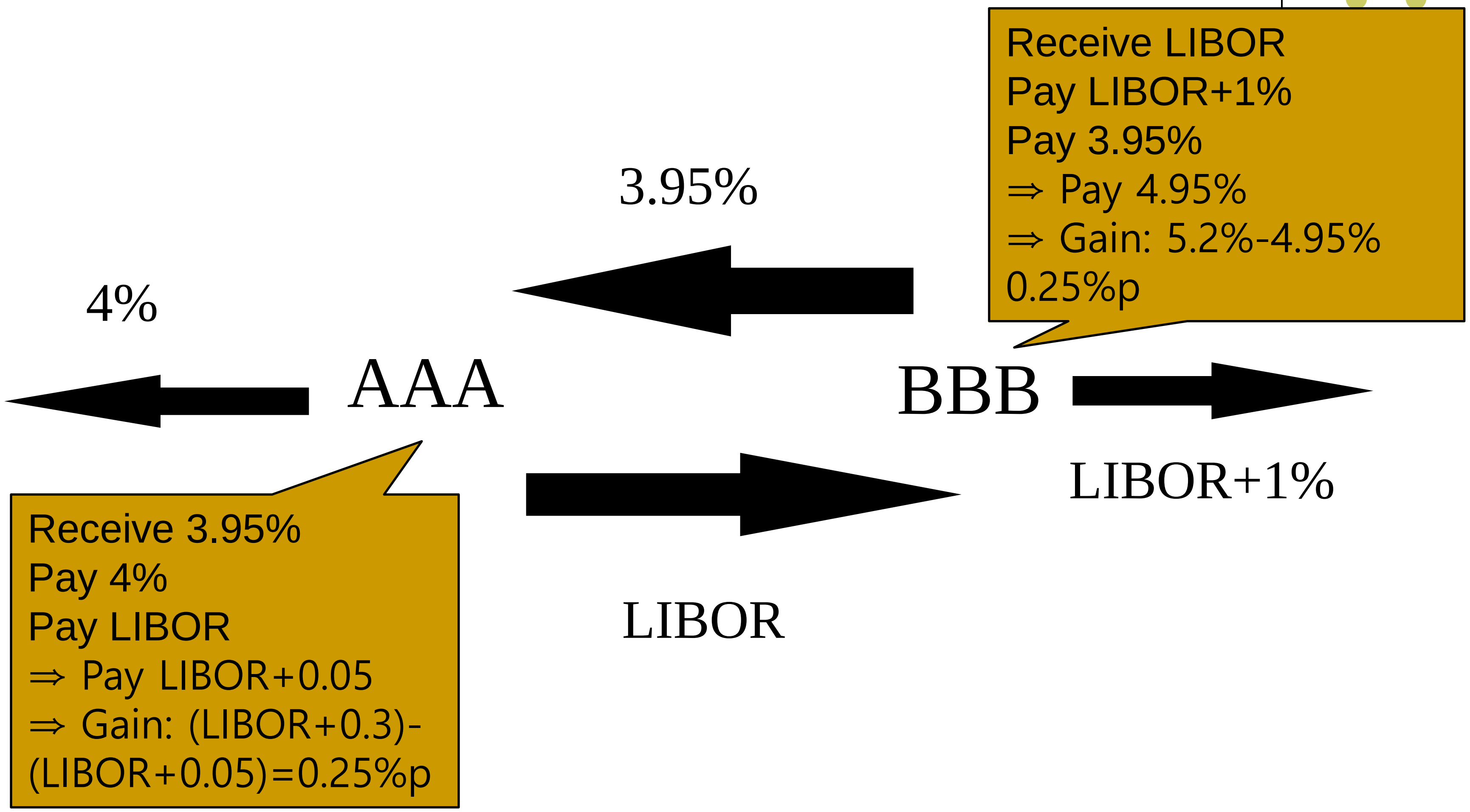


The Comparative Advantage Argument

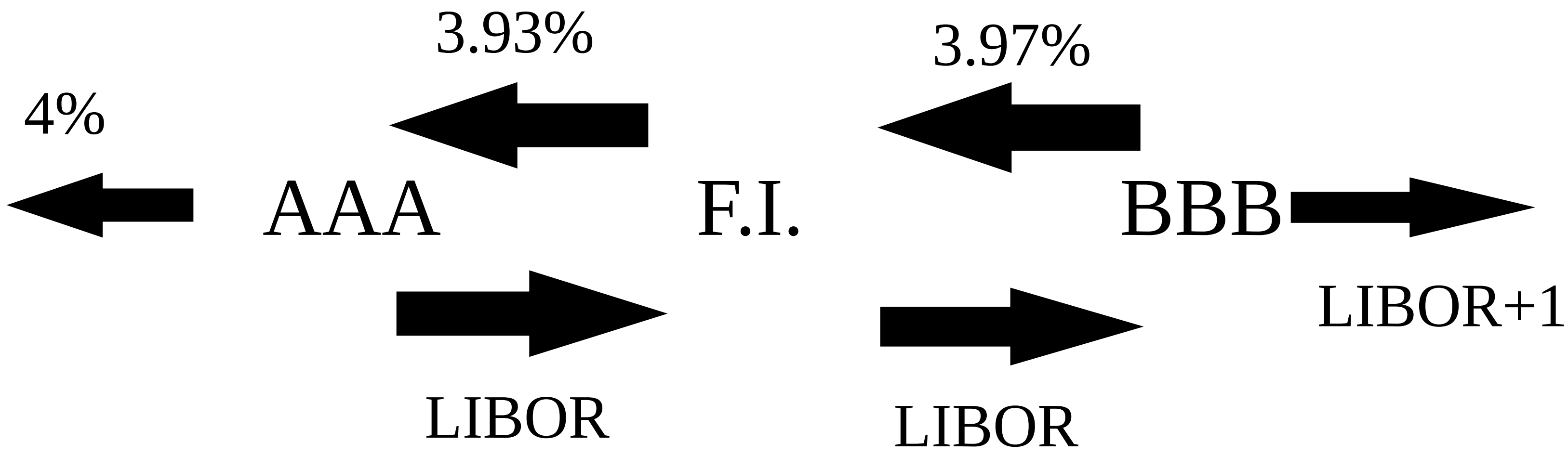
- AAACorp wants to borrow floating
- BBBCorp wants to borrow fixed

	Fixed	*Floating*
AAACorp	4.00%	6-month LIBOR + 0.30%
BBBCorp	5.20%	6-month LIBOR + 1.00%

The Swap


3.95%
4%
AAA
BBB
LIBOR
LIBOR+1%
Receive LIBOR
Pay LIBOR+1%
Pay 3.95%
⇒ Pay 4.95%
⇒ Gain: 5.2%-4.95% 0.25%p
Receive 3.95%
Pay 4%
Pay LIBOR
⇒ Pay LIBOR+0.05
⇒ Gain: (LIBOR+0.3)-(LIBOR+0.05)=0.25%p

The Swap when a Financial Institution is Involved



Criticism of the Comparative Advantage Argument

- The 4.0% and 5.2% rates available to AAACorp and BBBCorp in **fixed rate markets are 5-year rates**
- The LIBOR+0.3% and LIBOR+1% rates available in the **floating rate market are six-month rates**
- <u>BBBCorp's fixed rate depends on the spread above LIBOR</u> it borrows at in the future

The Nature of Swap Rates

- Six-month LIBOR is a short-term AA borrowing rate
- The 5-year swap rate has a risk corresponding to the situation where 10 six-month loans are made to AA borrowers at LIBOR
- This is because the lender can enter into a swap where income from the LIBOR loans is exchanged for the 5-year swap rate

Pricing on Swap Contracts

Swaps are essentially a series of forward contracts.

We need to find the level annuity, *F *,* with the same present value as the stream of annual cash flows that would be incurred in a sequence of forward rate agreements.

$$\frac{F_1}{(1+y_1)}+\frac{F_2}{(1+y_2)^2}=\frac{F^*}{(1+y_1)}+\frac{F^*}{(1+y_2)^2}$$

Forward Contracts versus Swaps

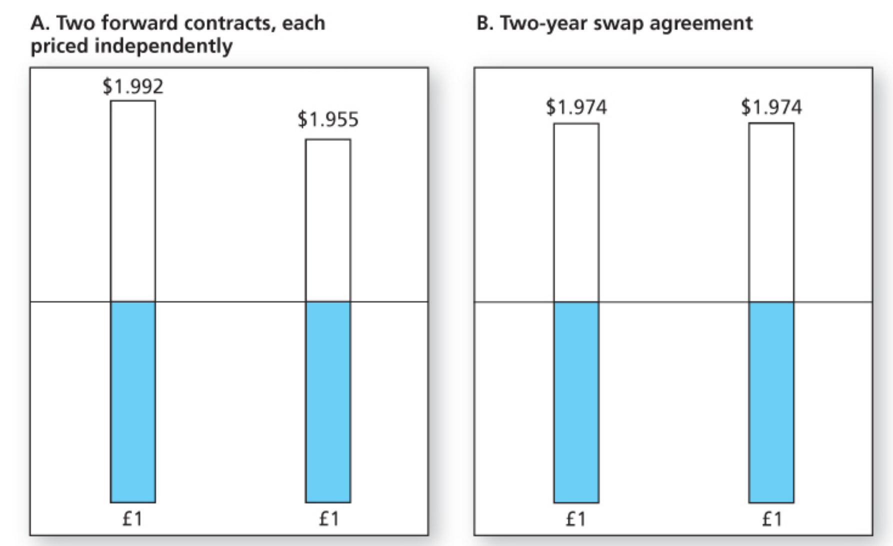


Figure 23.8 Forward contracts versus swaps

Valuation of an Interest Rate Swap

- Interest rate swaps can be valued as the difference between the value of a fixed-rate bond and the value of a floating-rate bond
- Alternatively, they can be valued as a **portfolio of forward rate agreements** (FRAs)

Valuation in Terms of Bonds

- The fixed rate bond is valued in the usual way
- The floating rate bond is valued by noting that it is worth par immediately after the next payment date

Example

- Receive 8% per annum and pay floating semiannually on a principal of $100 million.
- 1.25 years to go and next floating payment is $5.1 million

Time	Fixed Bond	Floating Bond	Disc Factor	PV fixed Bond	PV floating Bond
0.25	4	105.1	0.9753	3.901	102.5045
0.75	4		0.9243	3.697	
1.25	104		0.8715	90.64	
				98.238	102.505

- Swap value = 98.238 − 102.505= − 4.267

Valuation in Terms of FRAs

- Each exchange of payments in an interest rate swap is an FRA
- The FRAs can be valued on the assumption that today's forward rates are realized

Example

Time	Fixed Cash Flow	Floating Cash Flow	Net Cash Flow	Disc Factor	PV of Net Cash Flow
0.25	4	-5.100	-1.100	0.9753	-1.073
0.75	4	-5.522	-1.522	0.9243	-1.407
1.25	4	-6.051	-2.051	0.8715	-1.787
					-4.267

An Example of a Currency Swap

While the cost of borrowing in the international market is unreasonably high, companies have a competitive advantage for taking out loans from their domestic banks.

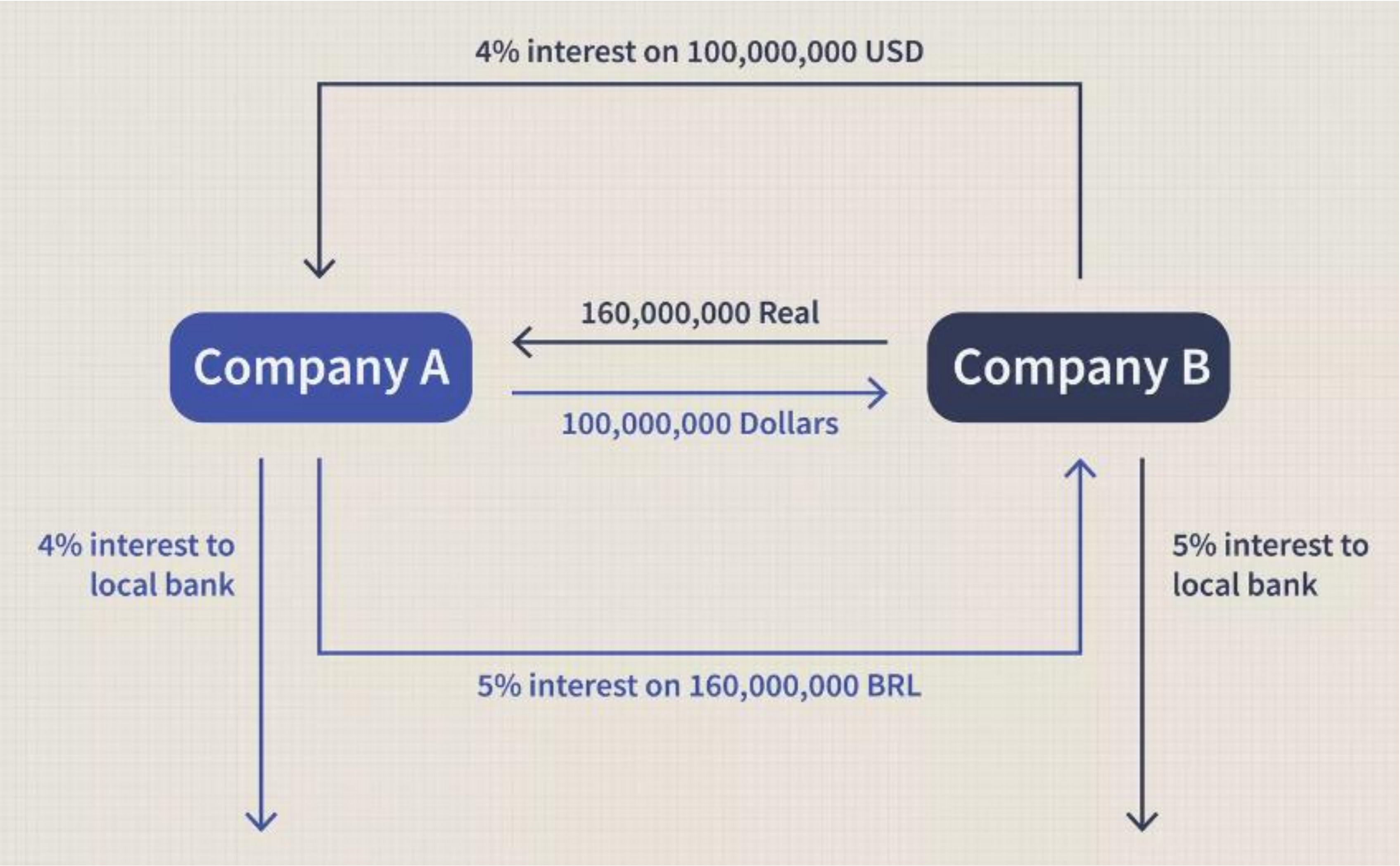
Company A could hypothetically take out a loan from an American bank at 4% and Company B can borrow from its local institutions at 5%.

Exchange of Principal

- In an interest rate swap the principal is not exchanged.

- the US$100 million and 160 million Brazilian real are exchanged when the contract is initiated. **At termination, the notional principals are returned** to the appropriate party.

The Cash Flows



Swap Dealer

- With the presence of the dealer, the realized interest rate might be increased slightly as a form of **commission** to the intermediary.

- Therefore, the actual borrowing rate for Companies A and B is 5.1% and 4.1%, respectively, which is still superior to the offered international rates: commission 0.1%

Typical Uses of a Currency Swap

- **Conversion from a liability** in one currency to a liability in another currency

- **Conversion from an investment** in one currency to an investment in another currency

Comparative Advantage Arguments for Currency Swaps

General Electric wants to borrow AUD

Qantas wants to borrow USD

	USD	AUD
General Motors	5.0%	7.6%
Qantas	7.0%	8.0%

Valuation of Currency Swaps

Like interest rate swaps, currency swaps can be valued either as the difference between 2 bonds or as a portfolio of forward contracts.

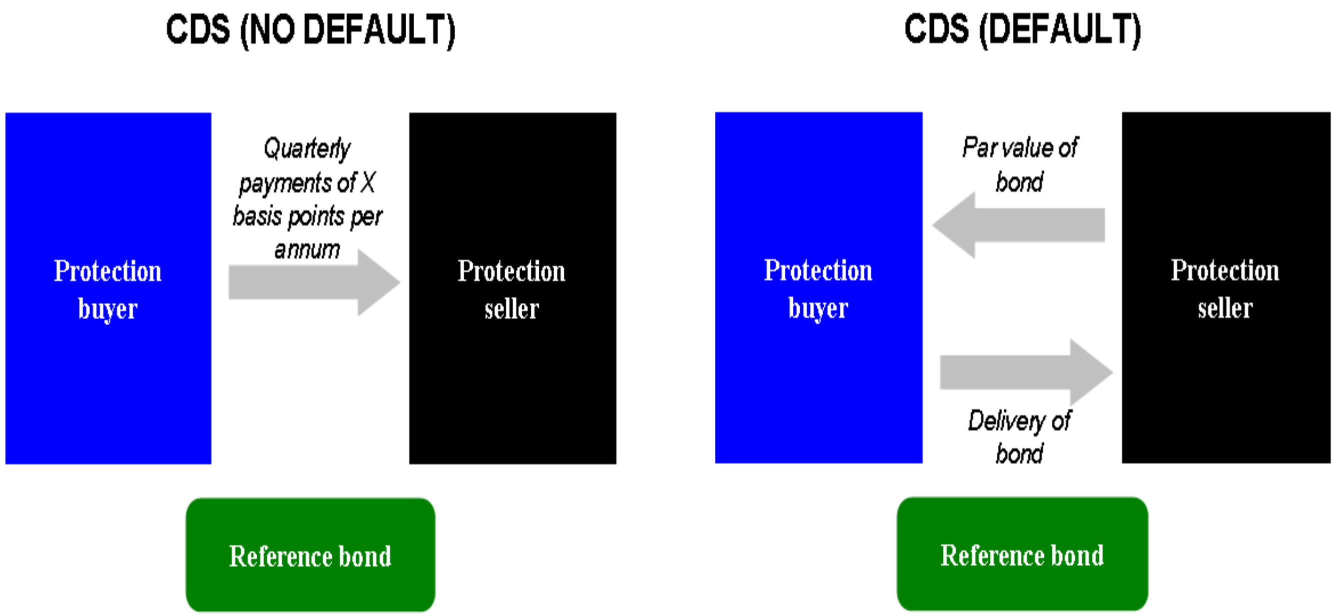
Swaps & Forwards

- **A swap can be regarded as a convenient way of packaging forward contracts**
- When a swap is initiated the swap has zero value, but typically some forwards have a positive value and some have a negative value

Credit Risk

- A swap is worth zero to a company initially
- At a future time its value is liable to be either positive or negative
- The company has credit risk exposure only when its value is positive

Credit Default Swaps



CDS Index: Example